# Health & Community Impacts of Data Centers

Six ways a data center affects the people who live near one — the health risks plus higher bills and the environment. Every claim sourced on the pages that follow, with the permit condition that addresses it.

## Air pollution

Fossil power plants and diesel backup generators serving data centers emit pollution linked to ~1,300 premature deaths a year by 2030.

## Noise pollution

Cooling systems and generators produce a constant low-frequency hum linked to sleep disturbance, hypertension, and cardiovascular disease.

## Light pollution

24/7 campus lighting disrupts sleep and circadian rhythms; the AMA has warned about nighttime light exposure since 2012.

## Higher bills

Large loads strain the grid and shift transmission and capacity costs onto residential ratepayers.

## Water consumption

Data centers consume water twice — on site for cooling and at the power plants that supply them — often in water-stressed regions.

## Climate & reliability

Fossil-heavy demand growth raises emissions, strains grid reliability, and adds local waste heat.

## AIR POLLUTION

- UC Riverside & Caltech researchers project up to ~1,300 premature deaths a year by 2030 from U.S. data-center air pollution — that is a midpoint, with a published range of 940 to 1,590 — alongside roughly 600,000 asthma-symptom cases, and a total public-health burden exceeding $20B in 2028. Quote the range rather than the midpoint: it is the defensible version, and a point estimate invites an argument about precision you do not need to have.

  *Source: Han, Wu, Li, Wierman & Ren — The Unpaid Toll: Quantifying the Public Health Impact of AI (arXiv:2412.06288, UC Riverside/Caltech, 2024)*
  https://arxiv.org/abs/2412.06288

- The pollution comes from two places: the fossil-fueled power plants supplying the electricity and on-site diesel backup generators emitting NOx and fine particulates (PM2.5) — both linked to respiratory disease, cardiovascular conditions, and cancer risk.

  *Source: Han, Wu, Li, Wierman & Ren — The Unpaid Toll: Quantifying the Public Health Impact of AI (arXiv:2412.06288, UC Riverside/Caltech, 2024)*
  https://arxiv.org/abs/2412.06288

- The burden lands unevenly: much of the health impact falls on communities near the power plants that serve a data center, which can sit hundreds of miles from the facility itself.

  *Source: Han, Wu, Li, Wierman & Ren — The Unpaid Toll: Quantifying the Public Health Impact of AI (arXiv:2412.06288, UC Riverside/Caltech, 2024)*
  https://arxiv.org/abs/2412.06288

### WHAT TO DEMAND

Demand generator run-hour limits, Tier 4 (or battery) backup instead of legacy diesel, and an air-permit review with public comment before any approval.

## NOISE POLLUTION

- Data-center cooling systems and backup generators produce a continuous low-frequency hum that travels farther than ordinary sound and penetrates building walls.

  *Source: Environmental Health Project — The Health Risks of Data Centers (infographic, 2026)*
  https://www.environmentalhealthproject.org/_files/ugd/a9ce25_3c3574ea12324c65909d308c3a716e56.pdf

- The WHO links chronic environmental noise to sleep disturbance, cardiovascular and metabolic disease, and cognitive impairment — in Europe, environmental noise contributes to an estimated 48,000 new heart-disease cases and 12,000 premature deaths every year.

  *Source: European Environment Agency — Health risks caused by environmental noise in Europe (2020)*
  https://www.eea.europa.eu/en/analysis/publications/health-risks-caused-by-environmental-noise-in-europe

- WHO night-noise recommendations sit around 40-45 dB — the reason our model CBA clause sets 45 dBA at the nearest residential property line.

  *Source: WHO Europe — Environmental Noise Guidelines: noise & cardiovascular/metabolic mechanisms (2018)*
  https://www.who.int/europe/publications/i/item/WHO-EURO-2018-3009-42767-59666

### WHAT TO DEMAND

A 45 dBA limit at the residential property line as a permit condition — measured after commissioning, not just modeled — with quarterly public reporting.

## LIGHT POLLUTION

- The American Medical Association adopted policy in 2012 recognizing nighttime light exposure as a health hazard: it suppresses melatonin and disrupts circadian rhythms.

  *Source: American Medical Association — Council on Science & Public Health report on light pollution & nighttime lighting (2012 policy)*
  https://www.ama-assn.org/sites/ama-assn.org/files/corp/media-browser/public/about-ama/councils/Council%20Reports/council-on-science-public-health/a12-csaph4-lightpollution-summary.pdf

- Chronic light at night is associated with reduced sleep, impaired daytime functioning, obesity, and mood disorders, with research linking elevated light pollution to higher breast and prostate cancer rates.

  *Source: American Medical Association — Council on Science & Public Health report on light pollution & nighttime lighting (2012 policy)*
  https://www.ama-assn.org/sites/ama-assn.org/files/corp/media-browser/public/about-ama/councils/Council%20Reports/council-on-science-public-health/a12-csaph4-lightpollution-summary.pdf

- Data center campuses typically run high-intensity exterior security lighting around the clock.

  *Source: Environmental Health Project — The Health Risks of Data Centers (infographic, 2026)*
  https://www.environmentalhealthproject.org/_files/ugd/a9ce25_3c3574ea12324c65909d308c3a716e56.pdf

**WHAT TO DEMAND**

Full-cutoff shielded fixtures, color temperature at or below 3000K (the AMA community guidance), and dark-sky-compliant site lighting as approval conditions.

## HIGHER BILLS

- PJM's 2025 capacity auction cleared at a record $16.4B, of which the market monitor attributes $6.3B — 38% — to data centers. Across PJM's last four base auctions it is $29.4B of $63.6B, or 46%. These costs flow through to ratepayers across 13 states.

  *Source: Monitoring Analytics (PJM market monitor) — data centers drove $6.3B of the record $16.4B 2025 capacity auction (38%), and $29.4B of $63.6B across the last four auctions (46%); see the 2025/2026 RPM Base Residual Auction analyses*
  https://www.monitoringanalytics.com/reports/Reports/2025.shtml

- Virginia's legislative audit agency (JLARC) found that unconstrained data-center growth will raise costs for other customers absent policy changes — in the state with more data centers than any other.

  *Source: Virginia JLARC — Data Center Impact Study (Report 591, Dec 2024)*
  https://jlarc.virginia.gov/pdfs/reports/Rpt591.pdf

- Berkeley economists warn that who pays for data-center-driven grid expansion is a policy choice: without large-load tariffs, the default is everyone.

  *Source: UC Berkeley Energy Institute — What Will Data Centers Do To Your Electric Bill? (2025)*
  https://energyathaas.wordpress.com/2025/09/29/what-will-data-centers-do-to-your-electric-bill/

### WHAT TO DEMAND

Cost causation as a condition: the developer pays 100% of interconnection and grid upgrades, under a large-load tariff so costs never reach residential bills. Estimate your own exposure in the Local Impact Calculator.

## WATER CONSUMPTION

- About one-fifth of data centers' direct water footprint is drawn from moderately-to-highly water-stressed watersheds, and nearly half of servers are powered by plants in water-stressed regions.

  *Source: Siddik, Shehabi & Marston — The environmental footprint of data centers in the United States (Environmental Research Letters, 2021)*
  https://iopscience.iop.org/article/10.1088/1748-9326/abfba1

- Water is consumed twice: directly for evaporative cooling on site, and indirectly at the thermoelectric power plants generating the electricity.

  *Source: Siddik, Shehabi & Marston — The environmental footprint of data centers in the United States (Environmental Research Letters, 2021)*
  https://iopscience.iop.org/article/10.1088/1748-9326/abfba1

- AI-specific demand is growing fast — researchers project AI's water withdrawal could reach billions of cubic meters a year by 2027, and most operators still don't report site-level water use.

  *Source: Li et al., Making AI Less Thirsty — water footprint of AI (arXiv:2304.03271)*
  https://arxiv.org/abs/2304.03271

### WHAT TO DEMAND

An enforceable annual water cap in the permit, recycled or non-potable cooling supply, quarterly public metering, and re-approval before expansion.

## CLIMATE & RELIABILITY

- U.S. data-center electricity use is projected to reach 325-580 TWh by 2028 (Berkeley Lab) — 6.7% to 12% of national consumption, up from 176 TWh and 4.4% in 2023 — with much of the new supply coming from natural gas.

  *Source: Lawrence Berkeley National Lab, 2024 US Data Center Energy Usage Report — 325–580 TWh (6.7–12% of US electricity) by 2028, up from 176 TWh (4.4%) in 2023*
  https://eta.lbl.gov/publications/2024-united-states-data-center-energy

- The IEA projects data centers will drive one of the largest sources of electricity demand growth this decade, with the fuel mix determining the emissions impact.

  *Source: IEA, Energy and AI (2025) + Key Questions update (2026)*
  https://www.iea.org/reports/energy-and-ai

- Hyperscalers are increasingly turning to on-site gas generation as a 'bridge' — locking in fossil combustion next to host communities.

  *Source: Natural Gas Intelligence — On-site natural gas generation gains favor with hyperscalers (2026)*
  https://naturalgasintel.com/news/on-site-natural-gas-generation-gains-favor-with-hyperscalers-as-bridge-to-grid/

### WHAT TO DEMAND

Binding 24/7 carbon-free energy commitments, no ratepayer-funded gas buildout, and demand-response/curtailment agreements so the facility sheds load in grid emergencies.